

Raphaël Vermeil

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EDUCATION

McGill University

May 2029

B.Eng. Software Engineering CO-OP | GPA: 3.8

Relevant Coursework: Product Management, Software Engineering and Design Process, Data Structures and Algorithms, Computer Systems and Organization, Data Science and Analytics, Communications

TECHNICAL SKILLS

Languages: Java, Python, C++, C, C#, JavaScript, TypeScript, SQL

Technologies: Docker, Git, Linux/Unix, React, Node.js, NumPy, DuckDB, Dagster, MongoDB, Socket.IO

PROFESSIONAL EXPERIENCE

Opteo | Software Engineering Intern

May 2026 - August 2026

- Built AI tooling by implementing **Model Context Protocol (MCP)** tools for internal **Fin/Intercom** assistants, enabling LLM-powered customer support agents to retrieve and act on operational account data.
- Engineered a scalable research data platform using **DuckDB** and **Dagster**, orchestrating self-healing data pipelines for **200,000+ advertising accounts** with **3+ years of historical campaign data**, enabling large-scale analytics and machine learning workflows.
- Developed machine learning forecasting pipelines using **Python**, **Prophet**, and **Gradient Boosting Models (GBM)** to predict advertising account performance and generate account health scores across hundreds of Microsoft and Google Ads accounts.

StudioXP | Programming Teacher

Nov 2020 - Feb 2024

- Instructed **130+ students** in object-oriented programming, software architecture fundamentals, and game development using **C#**, **Java**, and the **Unity** engine.
- Mentored students through software design, debugging, and code review, strengthening programming proficiency and independent problem-solving across **3+ years** of teaching.

PROJECTS

Neural Critters | Python, NumPy, NEAT, multiprocessing, pygame

- Built a custom **2D spring-mass physics engine** and artificial-life simulator (~2,100 LOC) that evolved user-designed creatures using **NEAT** neural networks.
- Improved simulation performance through **NumPy vectorization** and multicore parallelism, significantly reducing generation runtime

BridgeOrBust | Java, JavaFX, Maven

- Developed the core **physics engine** for a Java bridge-building game with **structural stress simulation**.
- Implemented a **resolution-independent physics** system for consistent gameplay.

Full-Stack Chat Application | React, Node.js, MongoDB

- Built a **real-time** full-stack chat application using React, Node.js, MongoDB, **Socket.IO**, and JWT authentication.
- Implemented secure **REST APIs**, live messaging, and online presence.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

- The Factory - Electronics Lab** | General Manager January 2026 - Present
- McGill Engineering Competition (Eng Games)** | VP Competition January 2026 - Present